CLE: Chess Learning Engine

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CMSI 4072: SENIOR PROJECT • LOYOLA MARYMOUNT UNIVERSITY

PROJECT OVERVIEW

The CLE is a proof-of-concept chess engine developed to explore the fundamentals of reinforcement learning.

Learning vs. Raw Power

Unlike professional engines like Stockfish, this project focuses on the learning mechanism rather than competitive strength. Currently, the engine successfully defeats random-move generators and Minimax depth 1 searches, though it still struggles against Minimax depth 2 opponents.

SYSTEM ARCHITECTURE

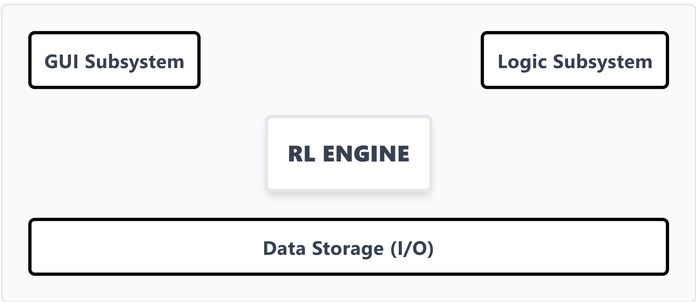

Figure 1: Modular CSCI Architecture

THE LEARNING LOOP

1. **Vectorization:** The board is converted into an 8x8x12 tensor representing piece locations.
2. **Inference:** The Neural Network performs a forward pass to calculate the winning probability.
3. **Backpropagation:** Post-game, the engine adjusts weights based on actual outcomes vs predictions.

KEY TECHNOLOGIES

- **Python 3.12:** Primary development language.
- **Pygame:** Renders the 8x8 visual grid and handles interactions.
- **TensorFlow/NumPy:** Manages matrix math and neural training.

VALIDATION & TESTING

- **Rule Compliance:** Verified FIDE move legality.
- **Baseline Testing:** Benchmarked against random moves and depth-limited Minimax.
- **Persistence:** Verified database I/O for weight storage.

CONCLUSIONS

The CLE demonstrates the successful implementation of a self-learning loop. While it does not yet compete with top-tier engines, it provides a valuable framework for understanding the intersection of game theory and neural networks.